

A unit-cube counterexample to the discrete covariogram Ball-body inclusion

Automated Math Discovery Run `fa_banach_001`
Model: GPT5.6

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Claimed status: counterexample, subject to human review. The normalized inclusion marked unknown in the source fails for $K = [0, 1]^n$, $p = 1$, and $q = 2$ in every dimension $n \geq 1$.

1 Source question

Alonso-Gutiérrez, Lucas, and Martín Goñi ask whether the normalized inclusion between continuous covariogram Ball bodies, their equation (2.16), also holds for the Ball bodies of the discrete covariogram [1, Section 6, p. 34]. The source passage is reproduced below.

Notice that, since $K \subseteq K + C_n$, then $\tilde{g}_K(x) \leq \tilde{g}_{K+C_n}(x)$ for every $x \in \mathbb{R}^n$. Thus, for every $\theta \in S^{n-1}$, we have

$$G_n(K)\rho_{K_p(\tilde{g}_K)}^p(\theta) \leq G_n(K + C_n)\rho_{K_p(\tilde{g}_{K+C_n})}^p(\theta)$$

and then

$$(6.2) \quad K_p(\tilde{g}_K) \subseteq \left(\frac{G_n(K + C_n)}{G_n(K)} \right)^{\frac{1}{p}} K_p(\tilde{g}_{K+C_n}).$$

The purpose of this section is to prove, on the one hand, that even though $K_p(\tilde{g}_K)$ is not necessarily convex, its convex hull is contained in the same dilation of $K_p(\tilde{g}_{K+C_n})$ that appears in (6.2) and, on the other hand, that even though we do not know whether an inclusion relation such as the one given by (2.16) holds for the p -th Ball bodies of the discrete covariogram, a similar inclusion relation holds when substituting $K_p(\tilde{g}_K)$ by the dilation of $K_p(\tilde{g}_{K+C_n})$ given by the right hand side of (6.2). More precisely, we will prove the following:

Theorem 6.2. *Let $K \subseteq \mathbb{R}^n$ be a convex body such that $0 \in K$. For any $0 < p < q$ we have that*

Figure 1: The unknown discrete inclusion in the source paper.

More explicitly, for every convex body $K \subseteq \mathbb{R}^n$ with $0 \in K$ and every $0 < p < q$, the candidate inclusion is

$$\binom{n+q}{n}^{1/q} K_q(\tilde{g}_K) \stackrel{?}{\subseteq} \binom{n+p}{n}^{1/p} K_p(\tilde{g}_K). \quad (1)$$

2 Definitions

Let $G_n(A) = |A \cap \mathbb{Z}^n|$. The discrete covariogram of K is

$$\tilde{g}_K(x) = G_n(K \cap (x + K)).$$

For $p > 0$, its p -th Ball body is the star set

$$K_p(\tilde{g}_K) = \left\{ x \in \mathbb{R}^n : p \int_0^\infty r^{p-1} \tilde{g}_K(rx) dr \geq \tilde{g}_K(0) \right\}.$$

Its radial function satisfies

$$\rho_{K_p(\tilde{g}_K)}(\theta)^p = \frac{p}{G_n(K)} \int_0^\infty r^{p-1} G_n(K \cap (r\theta + K)) dr. \quad (2)$$

3 Proof intuition

The lattice points of the unit cube lie in just two layers perpendicular to e_1 . At zero translation both layers contribute. As soon as the cube is translated a positive distance in direction e_1 , the layer with first coordinate zero disappears, while the layer with first coordinate one survives until translation distance one. Thus the discrete covariogram drops immediately from 2^n to 2^{n-1} and then remains constant. Every radial p -moment therefore has the same p -th power, namely $1/2$. The binomial normalizations at $p = 1$ and $q = 2$ then point in the wrong order.

4 Counterexample

Theorem 1. *For every integer $n \geq 1$, the convex body $K = [0, 1]^n$, with $p = 1$ and $q = 2$, violates (1). In particular, the proposed discrete analogue is false even in dimension one.*

Proof. The cube contains exactly 2^n lattice points, so $G_n(K) = \tilde{g}_K(0) = 2^n$. For $r > 0$,

$$K \cap (re_1 + K) = [r, 1] \times [0, 1]^{n-1} \quad \text{when } 0 < r \leq 1,$$

and the intersection is empty when $r > 1$. For $0 < r \leq 1$, its lattice points are precisely

$$\{1\} \times \{0, 1\}^{n-1},$$

so there are 2^{n-1} of them. Hence, for every $p > 0$, (2) gives

$$\rho_{K_p(\tilde{g}_K)}(e_1)^p = \frac{p}{2^n} \int_0^1 r^{p-1} 2^{n-1} dr = \frac{1}{2}. \quad (3)$$

The value of the integrand at $r = 0$ does not affect the integral.

Now take $p = 1$ and $q = 2$. The radial function in direction e_1 of the right-hand side of (1) equals

$$\binom{n+1}{n} \rho_{K_1(\tilde{g}_K)}(e_1) = \frac{n+1}{2}.$$

The corresponding radial function of the left-hand side is

$$\binom{n+2}{n}^{1/2} \rho_{K_2(\tilde{g}_K)}(e_1) = \frac{\sqrt{(n+1)(n+2)}}{2}.$$

The latter is strictly larger because $n+2 > n+1$. Inclusion of star sets would force the opposite inequality between radial functions in every direction. Therefore (1) fails. $\square$

5 Verification, scope, and novelty

The script `code/verify_unit_cube.py` uses exact rational arithmetic to enumerate the two lattice layers and check dimensions 1 through 20. It verifies that the squared excess of the proposed left radius over the right radius is exactly

$$\frac{(n+1)(n+2)}{4} - \frac{(n+1)^2}{4} = \frac{n+1}{4} > 0.$$

This computation is only a check; the proof above covers all dimensions.

The counterexample addresses exactly the undilated relation marked unknown. It does not contradict the source’s Theorem 6.2, which replaces $K_p(\tilde{g}_K)$ by a dilation of $K_p(\tilde{g}_{K+C_n})$ and includes an additional lattice-count factor.

The run’s lightweight indexes had no prior hit. A bounded search over arXiv, the published Cambridge version, and exact combinations of the title/id with “discrete covariogram”, “Ball bodies”, and “inclusion” found no later resolution or counterexample. This is not an exhaustive novelty claim. Human review should check the half-layer count in (3) and the two normalization constants.

References

- [1] D. Alonso-Gutiérrez, E. Lucas, and J. Martín Goñi, *A discrete approach to Zhang’s projection inequality*, Canadian Journal of Mathematics (2025), 1–47; arXiv:2509.14986. <https://doi.org/10.4153/S0008414X25101624>